

Baadhi — mapping flood damage from space

IIT Mandi Multimodal AI Hackathon 2026 · Track B · Case study: Trishuli glacial-lake outburst flood, 26 August 2026 (Copernicus EMSR927)

Given any area and a flood date, Baadhi builds a map of flood water and debris from Sentinel-1 radar and Sentinel-2 optical images, counts the buildings, roads and bridges in its path on the pre-event OpenStreetMap, and finds the settlements that lost their road link to a hospital or town — from raw data, in minutes on a laptop — and writes a one-page situation report. Its AI component is a flood segmentation network trained from scratch on Kuro Siwo: on live Sentinel-1 scenes of the 2025 Punjab floods it reaches a mean flood F1 of **0.77** against 0.52 for a radar threshold rule, and on the five Kuro Siwo test events it never saw a pooled F1 of **0.78** against 0.71 (it fails on one of them). On the Trishuli case study the flood/debris map agrees with the Copernicus EMS delineation at F1 0.90 and 0.92 (development areas) and **0.89 on the held-out area**.

1 • Problem and what the system answers

A glacial-lake outburst or monsoon flood destroys roads and bridges in narrow valleys within hours; responders need to know **where** the water and debris went, **what** it hit, and **who** is now cut off from care. The system must do this for an area and date chosen live by the judges, using only open satellite data and the map as it was before the event.

Question	Evidence	Output
Where did flood and debris hit?	Sentinel-1 before/after on every usable orbit track; Sentinel-2 when clear; flood model; DEM terrain rules	classes <i>flood water · debris/mud/scoured ground · affected channel</i> , confidence
What was damaged?	overlay on OpenStreetMap at <i>event – 2 days, 00:00 UTC</i>	buildings hit / at the edge, blocked road stretches, bridges over an affected reach, health facilities
Who is cut off?	road graph before vs after (network 25 km beyond the area), walking paths separately	per settlement: minutes to the nearest hospital/clinic and town, before and after
Bonus: where would a flood go?	steepest descent on Copernicus DEM, continued across windows	path and settlements, km downstream, height above the river

2 • Data and the rules we keep

Inputs: Sentinel-1 RTC (γ^0), Sentinel-2 L2A and Copernicus DEM GLO-30/90 from Microsoft Planetary Computer; OpenStreetMap as it was before the event (Overpass "attic" query, two days before the event date at 00:00 UTC — EMSR927 began on 25 Aug 22:00 UTC; if that server is slow, the newest dated Geofabrik regional snapshot from before the event, cut out locally); Kuro Siwo for training only. **Copernicus EMS, UNOSAT and other published flood maps are never inputs:** EMSR927 and EMSR838 are used only to score results, in code under experiments/ that the system never imports. No detector threshold was fitted to EMS maps (section 7).

3 • Method

3.1 Flood and debris map

- **Radar change** — for each track, up to three before-images and the first after-image; 3×3 multilook; log-ratio divided by each pixel's own pass-to-pass variability ($z \geq 3$, or two tracks ≥ 2). Layover, shadow and strong foreshortening are masked per track from the DEM and the satellite's heading and incidence.
- **Radar water** — very dark after ($VV \leq -15$ dB, $VH \leq -22$ dB) and ≥ 3 dB darker than before.
- **Optical** — clear pixels only (scene-classification mask), per-date clear-fraction selection, first-clear-pixel composites: vegetation stripped (NDVI -0.20), fresh sediment (bare-soil or brightness rise with NDVI loss), new water (MNDWI). Where Sentinel-2 saw the ground clearly and unchanged, radar change alone does not count (farmland harvests make radar noisy).
- **Terrain** — height above the main river (HAND, pysheds on the DEM with OSM rivers burned in) and slope gate what can be water (≤ 30 m) or debris (≤ 80 m; steep scoured banks allowed within 60 m of the river). Zones must connect to the river corridor; valley floor below the observed flood level is filled where clouds or radar shadow hid the ground.
- **Flood model** — water the network finds (probability ≥ 0.5) enters directly and is not forced to touch the corridor (on plains, flood water breaks into patches between dry fields).

3.2 AI component: flood segmentation model

U-Net with a ResNet-18 encoder (12.5 M parameters), **trained from scratch** on Kuro Siwo only — no ImageNet weights, so no data outside the allowed lists — on a laptop CPU. Input: 8 channels (VV and VH after and before, their differences, slope, height above the local valley floor); output: no water / permanent water / flood water. Official Kuro Siwo event splits: 23 training events, 11 validation events (four training events from different climates moved to validation), **official test events never seen** (including Nepal, 1111007). Loss: weighted cross-entropy + Dice on flood; random crops, rotations, flips, independent calibration offsets on the before/after images; weight averaging (EMA).

Making live data look like the training data. Kuro Siwo is σ^0 with a Lee filter; Planetary Computer gives terrain-flattened γ^0 without filtering. Live images are converted ($\sigma^0 \approx \gamma^0 \cdot \cos \theta$, incidence from the swath geometry) and Lee-filtered, then the *same* feature code as in training runs (baadhi/mL/features.py). Inference uses 512-pixel windows blended at the seams, averaged with the mirror image; per pixel,

the highest probability among tracks that see it properly. The model is exported to ONNX and run with onnxruntime (identical output, 1.6× faster than PyTorch; the dashboard does not need PyTorch).

3.3 Damage, access and flood path

Buildings are rasterised on the 10 m grid: ≥ 50 % inside the footprint = *hit*, touching it = *at the edge*. A road is *blocked* where ≥ 20 m of it runs continuously under water or debris; bridges within 30 m of an affected reach are *at risk*. Access: an undirected road graph with class speeds (trunk 50 km/h ... track 12 km/h), multi-source Dijkstra from all hospitals/clinics and towns (including those up to 25 km outside the area) before and after removing blocked stretches and at-risk bridges; a settlement is *cut off* if it loses its route, a *long detour* if the time doubles and grows ≥ 30 min. Footpaths are walked at 3 km/h to tell whether a cut-off village can still be reached on foot. The flood path follows D8 directions on GLO-90 and lists named places ≤ 400 m from the path and ≤ 60 m above the river.

4 · Results — Trishuli GLOF vs Copernicus EMSR927

EMSR927 areas 1–3 were used while developing the detector; **area 5 (Phosretar) was held out** of its tuning. The scores are for the deployed system — physical rules plus the flood model, which never saw EMSR927 or any other EMS map — with OpenStreetMap as of 24 Aug 2026, computed inside each EMS area of interest and excluding what EMS marked "not analysed".

Area	Precision	Recall	F1	IoU	EMS damaged buildings we flag	Damaged road length in footprint	Damaged bridges flagged / in OSM
AOI 1–2 Syabru Bensi– Timure dev	0.99	0.82	0.90	0.82	94 %	87 %	5 / 5
AOI 3 Bidur dev	0.95	0.90	0.92	0.86	95 %	85 %	20 / 20
AOI 5 Phosretar held out	0.86	0.93	0.89	0.81	80 %	77 %	30 / 30

The rule-only detector (no flood model) scored F1 0.89 and 0.92 on the development areas and — on Phosretar, which we looked at in that form first — 0.86 (precision 0.97, recall 0.77; buildings 63 %, roads 55 %, bridges 24 / 30). The flood model adds recall in the wide, braided reach at Phosretar at some cost in precision. Without any OpenStreetMap (rivers taken from the terrain only) the map still scores F1 0.89 at Bidur and 0.90 at Phosretar, but counts no buildings, roads or settlements.

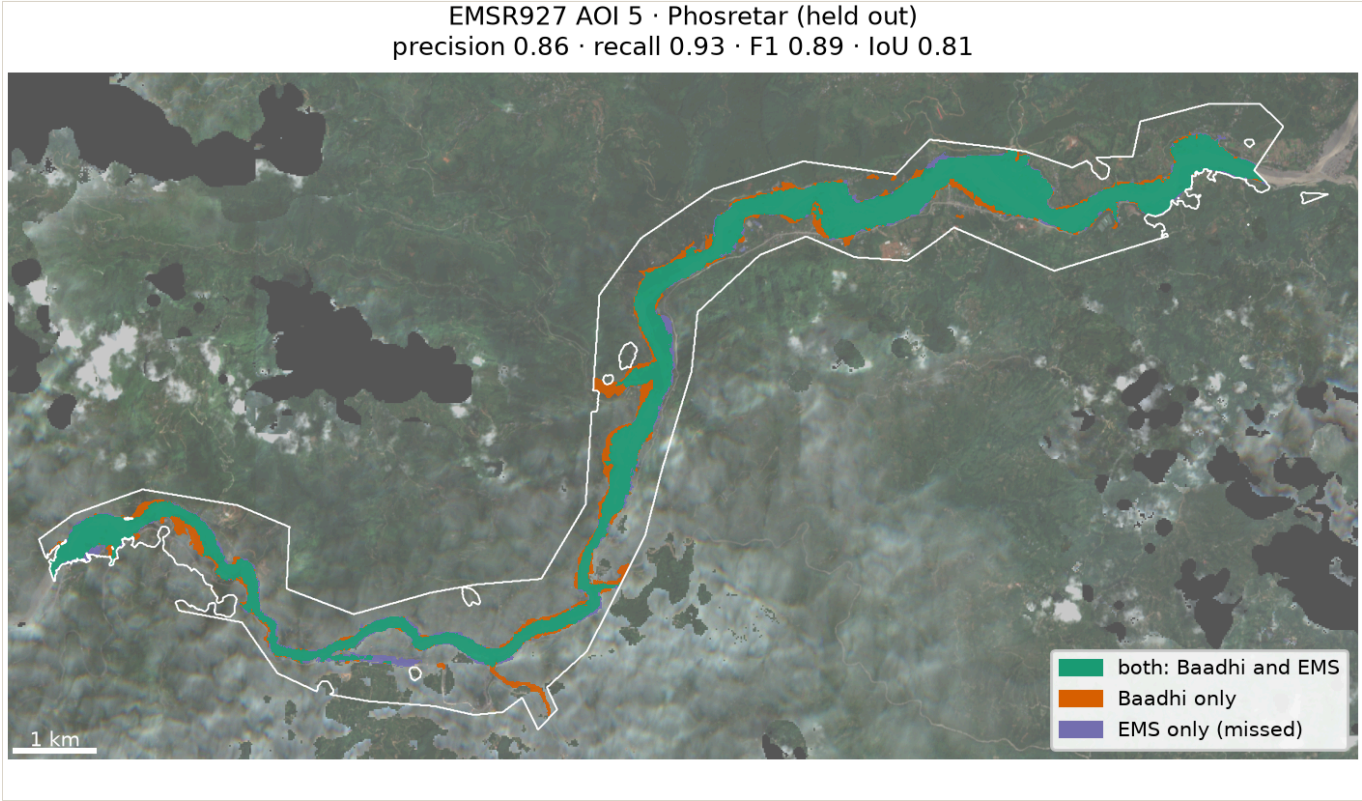


Figure 1. Held-out area, deployed system: green = both Baadhi and EMS, orange = Baadhi only, purple = EMS only. Baadhi marks more of the banks than EMS's delineation (precision 0.86) and misses little (recall 0.93).

Buildings: in the upper valley EMS graded 760 buildings damaged or destroyed; 94 % of them lie in or at the edge of our footprint. "Precision" of building flags (our hit buildings with an EMS-graded point within 15 m) is 82 % upper, 86 % Bidur and 52 % Phosretar — a rough proxy, since EMS graded only what it could see in very-high-resolution images. **Access:** in the upper valley 22 settlements lose their road link to the nearest hospital or town — Syabru Bensi had a clinic 24 minutes away before the event and none by road after.

Checking the cut-off analysis. The same routing was run twice on the same pre-event road network: once with roads cut by *our* footprint, once with roads cut where *EMS* graded them destroyed or damaged (both limited to the EMS areas). Every settlement's status (cut off / long detour / connected) was compared:

Area	Same status	Cut off: ours / EMS-based
AOI 1–2 Syabru Bensi–Timure dev	30 / 30 (100 %)	22 / 22
AOI 3 Bidur dev	89 / 89 (100 %)	5 / 5 (and 33 / 33 long detours)
AOI 5 Phosretar held out	93 / 101 (92 %)	13 (+6 "possibly") / 17 (11 in both)

In the held-out area EMS-based routing cuts off 17 settlements: we call 11 of them cut off and flag the other 6 as "possibly cut off — verify" (all 17 are flagged one way or the other), and EMS-based routing still finds a road for 2 settlements that we call cut off. 23 % of the road length EMS graded damaged lies outside our footprint.

5 • Results — the flood model on data it never saw

Which checkpoint. The rule, fixed before any evaluation, was to deploy the final epoch (30 of 30, averaged weights). Validation flood F1 peaks early (0.662 at epoch 2) and then stays within 0.647–0.662, while the IoU of "any water after the event" keeps rising (0.65 → 0.71) and so does the IoU of the permanent-water class (0.29 → 0.36), so training was not stopped early. The best-validation checkpoint (epoch 2) is shown for comparison.

Kuro Siwo official test events (never seen; 8,450 tiles from five events) — flood F1:

Event	Tiles	Rule	Model	Epoch 2
561	38	0.74	0.89	0.88
562	67	0.66	0.39	0.43
1111002	27	0.49	0.79	0.67
1111007 Nepal (Koshi plains)	2,318	0.63	0.78	0.76
1111013 USA	6,000	0.77	0.80	0.71
All, pooled	8,450	0.71	0.78	0.73

On event 562 the model labels 70 % of the flood pixels as *permanent* water (the before-images already showed water), so it misses most of the flood: F1 0.39 against 0.66 for the rule. The rule is scored with permanent water counted as "not flood", the easier case for it.

Live Sentinel-1, EMSR838 Punjab floods (Aug 2025). For each EMS area we take the ~14 km box with the most mapped flooding, use exactly the Sentinel-1 pass EMS mapped from, before-images from before the flood began, and score inside the EMS area only.

EMSR838 area	EMS flood	Rule F1	Model F1	Epoch 2
AOI01 Rasoo Nagar	143 km ²	0.63	0.93	0.95
AOI02 Sharaqpur	65 km ²	0.31	0.89	0.91
AOI04 Najabat	26 km ²	0.85	0.88	0.90
AOI05 Trimmu	9 km ²	0.30	0.39	0.46
Mean		0.52	0.77	0.80

Trimmu is a braided river: 317 flood patches with a median of 80 m², many at the scale of a pixel; the monsoon before-images already held water in many braids, which the model then calls permanent rather than new flood water.

6 • Dashboard, report and speed

The dashboard (FastAPI + MapLibre, python `serve.py`) takes an area drawn on the map and a date, streams progress, and shows the classes, confidence and model probability, Sentinel-2 before/after and a radar change composite, buildings, blocked roads, bridges, health facilities and settlements by access status; clicking a settlement shows its travel times. Each run produces the one-page situation report (HTML/PDF, every number from the maps), GeoTIFFs and GeoJSON layers. Timings measured on this laptop (10-core CPU, no GPU), pipeline time without the report (about 10–15 s more): a new 16 × 14 km area (Silchar 2022) took **221 s** from a cold start — radar and optical downloads about 2 min, pre-event OpenStreetMap 115 s from the public history server, the rest compute. With the downloads cached the case-study areas run in 70 s (upper valley), 106 s (Bidur) and 92 s (Phosretar); Wayanad 2024 in 108 s. In dense areas the Overpass history server is the bottleneck (Bidur 638 s, Phosretar 876 s measured, even with requests hedged over its three official instances); after 45 s without an answer the system therefore cuts the area out of a Geofabrik snapshot from before the event in three low-priority worker processes (275 s for a 55 × 55 km cell the first time, instant afterwards) and uses whichever source answers first. If neither answers in 10 minutes it still delivers the flood map and says that damage and access were not computed.

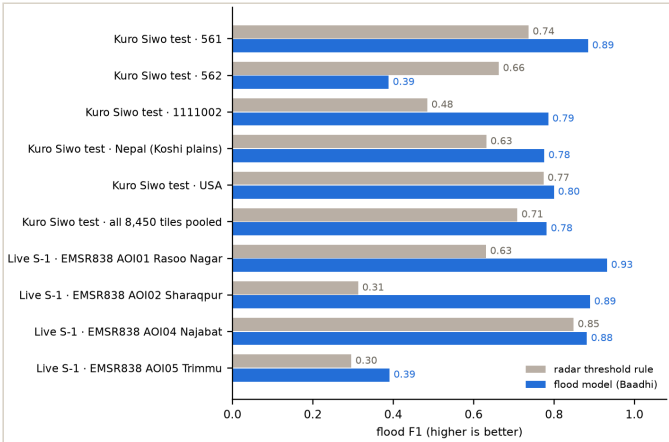


Figure 2. Flood F1 of the model (final epoch) and of the radar threshold rule on scenes never used in training — including the one test event the model loses.

7 · Limitations and honesty

- **Revisit gaps:** 6–12 days per track; water that drained before the first pass after the event is not seen; nothing is predicted in advance.
- **Radar blind spots** on steep slopes facing towards/away from the satellite; optical images need clear sky (rare in the monsoon).
- **"Hit" is not "destroyed":** at 10 m a building inside the footprint may stand; EMS grades damage from very-high-resolution imagery.
- **OpenStreetMap completeness:** unmapped buildings, bridges and villages are not counted — 9 of the 39 bridges EMS graded at Phosretar are not in OSM.
- **Travel times** use typical class speeds; roads outside the analysed area are assumed passable.
- **Where things were tuned:** detector thresholds are physical and fixed; they were checked on EMSR927 areas 1–3 (development). Area 5 was held out of that tuning: the rule-only detector was scored there first and the deployed system later — both are reported and nothing was chosen from them. The flood model never saw EMS data; on EMSR838 it is an independent test, while two plains rules were added after inspecting AOI01 (model water need not touch the river corridor; unknown height-above-river accepted on flat ground), so the combined detector's EMSR838 scores (mean F1 0.78) are a development check.
- **The model can mistake new flood water for permanent water** (Kuro Siwo event 562: F1 0.39 against 0.66 for the plain rule); where the before-images already show water, treat the mapped flood extent as a lower bound.
- **Map-server dependence:** the pre-event OpenStreetMap comes from a free public history server that needs 10–15 minutes in dense areas. The fallback is the newest dated Geofabrik snapshot from before the event (named in the report; for events before 2025 it can be up to a year older); without either, damage and access are not computed.
- **Kuro Siwo's Nepal event lies in the Koshi plains;** mountain performance is shown by the EMSR927 case study.
- Educational prototype, not an operational tool. No images of victims are used.

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